Cabin - Ping DFD

Leon Starr
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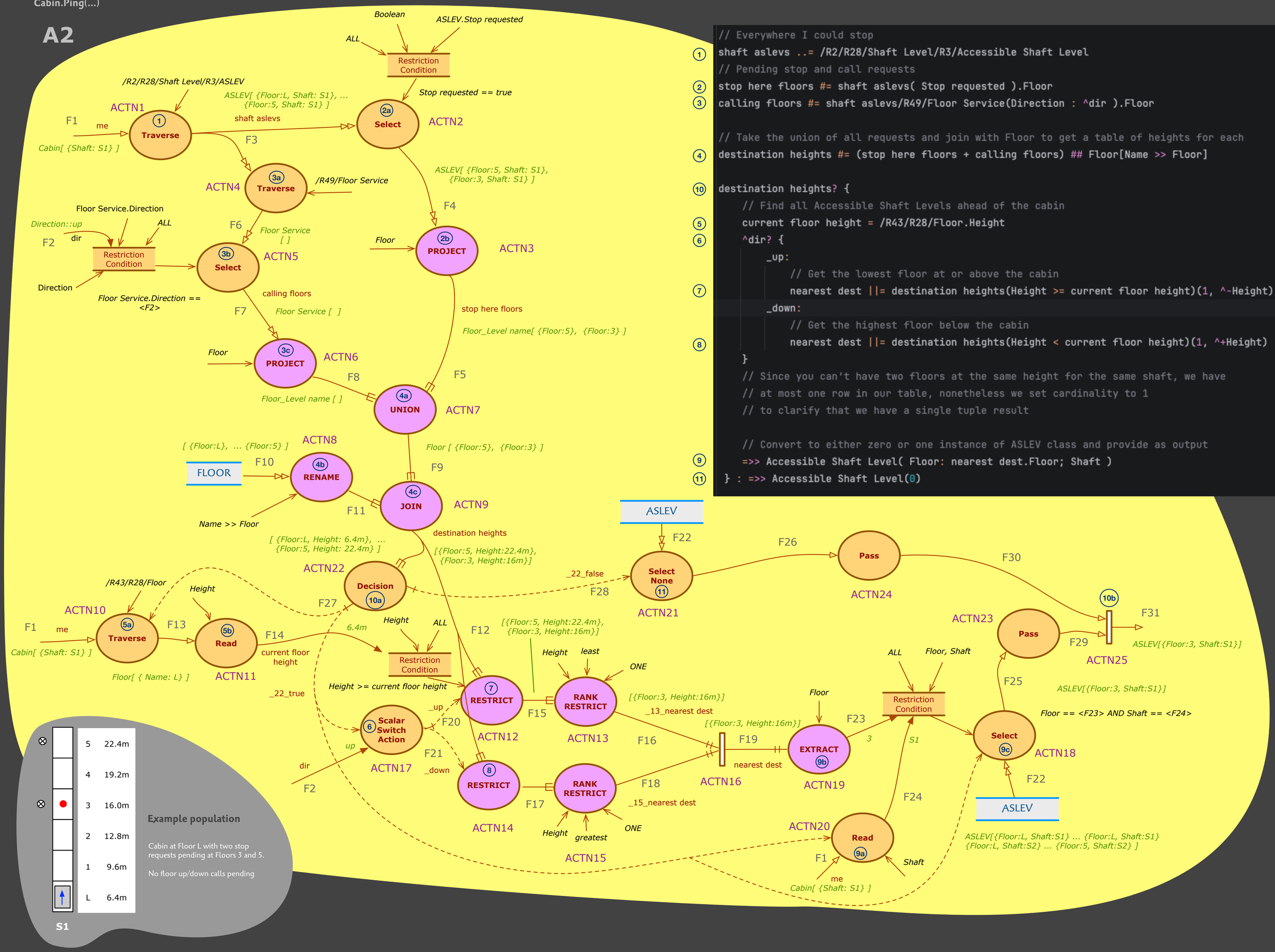
A2




```
// Everywhere I could stop
shaft aslevs ..= /R2/R28/Shaft Level/R3/Accessible Shaft Level
// Pending stop and call requests
stop here floors #= shaft aslevs( Stop requested ).Floor
calling floors #= shaft aslevs/R49/Floor Service(Direction : ^dir ).Floor

// Take the union of all requests and join with Floor to get a table of heights for each
destination heights #= (stop here floors + calling floors) ## Floor[Name >> Floor]

destination heights? {
    // Find all Accessible Shaft Levels ahead of the cabin
    current floor height = /R43/R28/Floor.Height
    ^dir? {
        _up:
            // Get the lowest floor at or above the cabin
            nearest dest ||= destination heights(Height >= current floor height)(1, ^-Height)
        _down:
            // Get the highest floor below the cabin
            nearest dest ||= destination heights(Height < current floor height)(1, ^+Height)
    }
    // Since you can't have two floors at the same height for the same shaft, we have
    // at most one row in our table, nonetheless we set cardinality to 1
    // to clarify that we have a single tuple result

    // Convert to either zero or one instance of ASLEV class and provide as output
    =>> Accessible Shaft Level( Floor: nearest dest.Floor; Shaft )
} : =>> Accessible Shaft Level(0)
```

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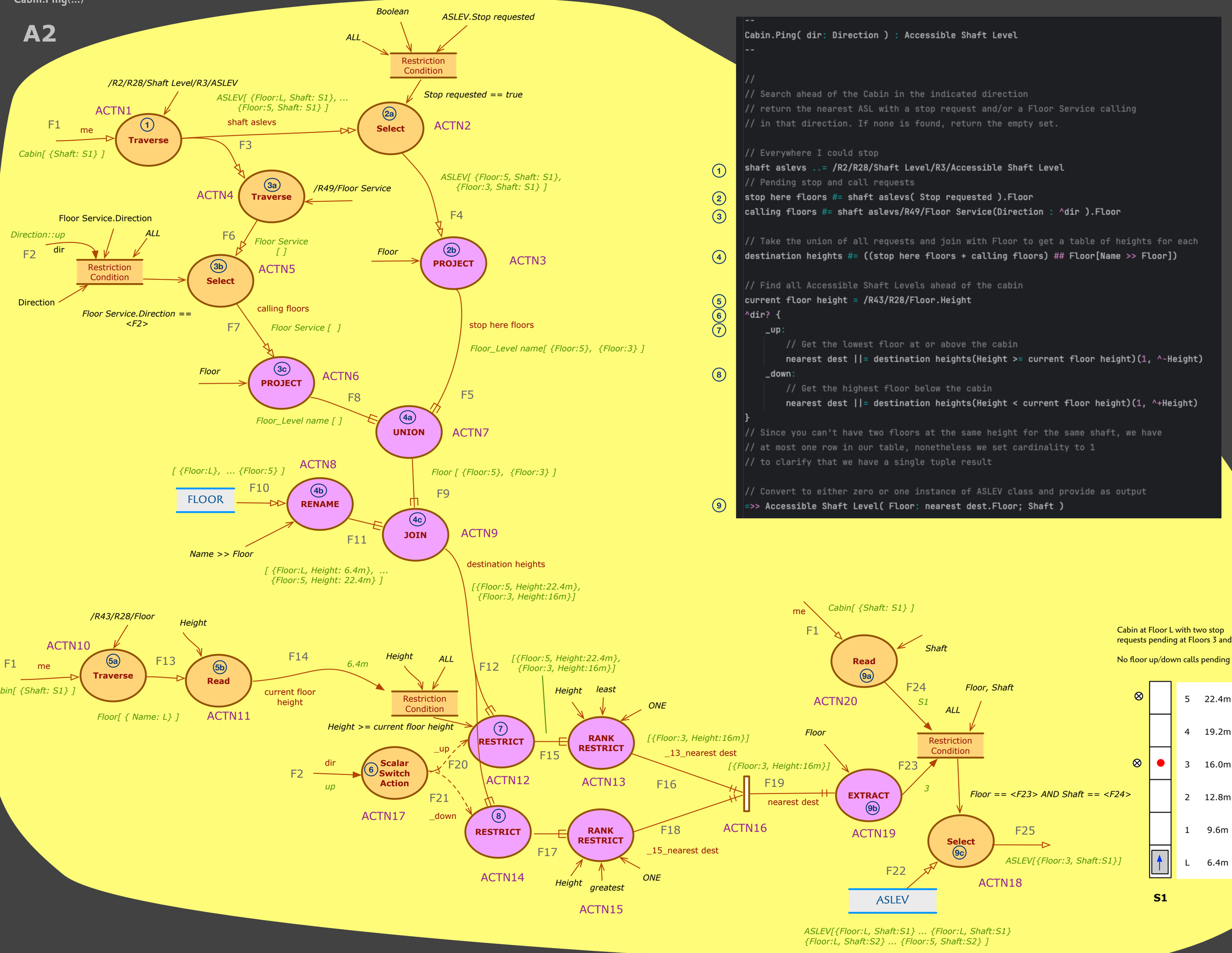


```
--
Cabin.Ping( dir: Direction ) : Accessible Shaft Level
--

//
// Search ahead of the Cabin in the indicated direction
// return the nearest ASL with a stop request and/or a Floor Service calling
// in that direction. If none is found, return the empty set.

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shaft aslevs ..= /R2/R28/Shaft Level/R3/Accessible Shaft Level
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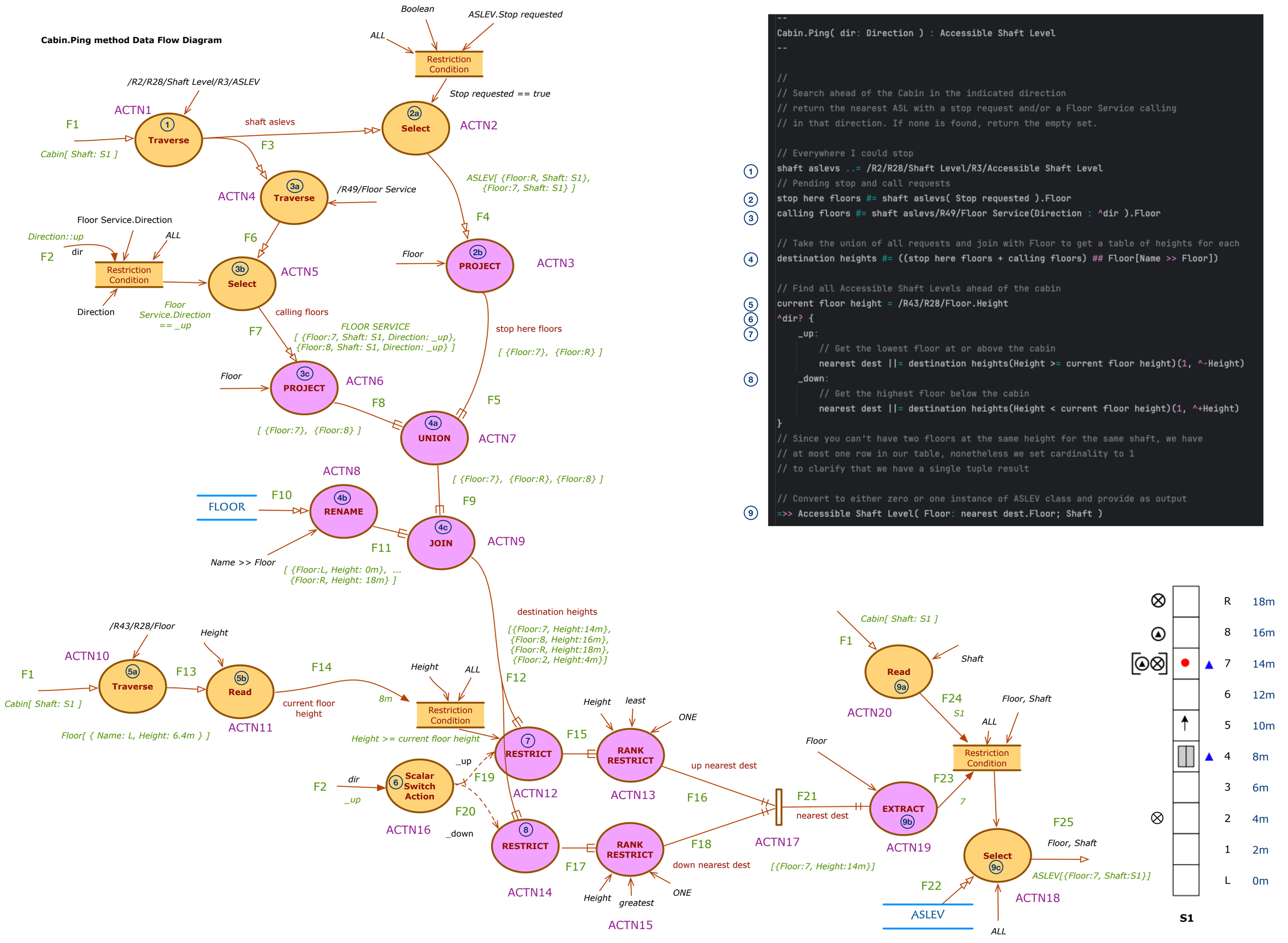
Cabin.Ping method Data Flow Diagram

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